

Name: Viechesta Layritz
Date: Mar 8, 2025
Title: Primary Research Plan

Research Timeline

- **Week of 3/2:** Data Collection and Coding Data
- **Week of 3/9:** Spring Break and Data Collection and Coding Data
- **Week 3/16:** Analyzing Data and Primary Data Report (class meets online)

Research Overview

1. **What do you intend to research?** (Provide a brief description of your research topic.)

The topic of my research paper is “The Role of Machine Learning in the Data Science Community.” Through this research, I intend to examine the utilization of machine learning in my discourse community. It includes how prevalent machine learning is in this community, algorithms that are commonly used in the data science community, and the most popular topics discussed among members in this community. The first half of the paper contains the basic definitions of terms that will be discussed further in the second half of the paper, which consists of assessing data science and machine learning through actual conversations from this community. The dataset that will be used in the analysis chapter is taken from the Stack Exchange forum. It is then processed in R Studio for data cleaning and further analysis. After going through the data massaging step, the dataset contains around 600 observational units.

2. **What is your research question(s)?** (Clearly state the main question(s) guiding your research.)

The main research question of my paper is “The Role of Machine Learning in the Data Science Community.” The study aims to understand the importance of machine learning in the data science community. The more important this field is, the more imperative it becomes for data scientists to acquire expertise in machine learning.

Primary Research Methods

Select 1 method from the list below:

- **Interview (1–3 people)**

- Observation (1 hour)
- Survey (10–15 people)
- Text Analysis (content or discourse)

Method 1:

- Which method are you using?

I plan to use the text analysis method for my research.

- Why did you choose this method?

I chose this method since this method allows me to have more individuals whom I can observe and make summaries of. Additionally, I am able to demonstrate an example of data science applications in processing data through incorporating this method in my research.

- How will you conduct this research? (Provide a step-by-step plan.)

First, I will collect the data from the Stack Exchange community forum about the data science community. Then, I will clean the data, which involves handling missing values. Subsequently, I will explore the data through graphical and numerical summaries. Next, I will interpret the findings from my visualizations.

- What challenges do you anticipate? (Consider potential obstacles.)

The challenges will be the time taken for me to realize all my ideas into a tangible action and the relevance of my data source since as a consequence of the development of AI. People might not see online community forums as the first thing that comes to mind whenever they have a question.

Method Guides

******Use the one relevant to your research and delete the others******

Fill in the BLUE highlighted areas BEFORE your primary research.

Fill in the YELLOW highlighted areas AFTER your primary research.

Text Analysis Guide

Component

Details

Texts Selected

List the 3–5 texts you will analyze.

	Question.Id	Title	Score	View.Count	Answer.Count	Owner.DisplayName	Creation.Date	Link
1	19048	What is the difference between test set and validation...	570	67508	12	xiaohu2012	2011-11-28	https://stats.stackexchange.com/questions/19048/
2	133656	How to understand the drawbacks of K-means	448	180054	5	KevinKim	2015-01-16	https://stats.stackexchange.com/questions/133656/
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Top 5 by Score

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4	120764	How does an LLM Agent parameter relate to a...	18	19966	1	slim_wizard	2023-04-06	https://datascience.stackexchange.com/questions/1...
5	590861	What is the difference between something being Aqu...	33	5745	5	stochasticmfox	2022-10-02	https://stats.stackexchange.com/questions/590861/...

Top 5 by Date

Purpose of Analysis

What are you looking for?

- *Examining how machine learning concepts are discussed in a public forum.*
- *Assessing areas where members in this community struggle the most.*
- *Identifying revealed patterns about machine learning in this community.*

Key Features

Themes, language, structure, argumentation, etc.

- *Model Evaluation*
- *Algorithm Strengths and Limitations*
- *High-Dimensional Data*
- *Statistical Terminologies*
- *Machine Learning Languages*
- *Semi-Formal*

Comparative Analysis

How do the texts differ in approach?

Questions in the subset top 5 by score and top 5 by date are different in approach since they ask about different steps in the data science workflow (data cleaning, data wrangling, exploratory data analysis, feature engineering, predictive modeling, and data visualization).

Component	Details
Examples & Evidence	<i>Quotations, citations, key excerpts.</i> • <i>“Why is Euclidean distance not a good metric in high dimensions?”</i> • <i>“What is the difference between test set and validation set?”</i> • <i>“How to understand the drawbacks of K-means”</i> • <i>“How to know that your machine learning problem is hopeless?”</i> • <i>“ROC vs precision-and-recall curves”</i> • <i>“Use of training data that has been labeled by the AI model itself”</i> • <i>“In training a neural network, why don’t we take the derivative with respect to the step size in gradient descent?”</i> • <i>“Why does data science see class imbalance as a problem for supervised learning when statistics does not?”</i> • <i>“How does an LLM ‘parameter’ relate to a ‘weight’ in a neural network?”</i> • <i>“What is the difference between something being ‘true’ and ‘true with probability 1’?”</i>

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Findings & Interpretation	<i>What do these texts reveal about your topic?</i> <i>Based on the selected texts:</i> • <i>Conceptual questions were still asked oftentimes.</i> • <i>Model Evaluation is one of the major concerns.</i> • <i>Interpretations matter.</i> • <i>Revisions are sometimes needed to ensure models can perform well on unseen datasets.</i> <i>According to the whole 626 observational units:</i> • <i>Machine learning is the most discussed topic among this discourse community.</i> • <i>After 2016, the number of active members in this community decreased.</i> Questions per year <table border="1"> <caption>Questions per year</caption> <thead> <tr> <th>Year</th> <th>Count</th> </tr> </thead> <tbody> <tr><td>2010</td><td>10</td></tr> <tr><td>2011</td><td>30</td></tr> <tr><td>2012</td><td>40</td></tr> <tr><td>2013</td><td>35</td></tr> <tr><td>2014</td><td>75</td></tr> <tr><td>2015</td><td>75</td></tr> <tr><td>2016</td><td>93</td></tr> <tr><td>2017</td><td>90</td></tr> <tr><td>2018</td><td>90</td></tr> <tr><td>2019</td><td>35</td></tr> <tr><td>2020</td><td>30</td></tr> <tr><td>2021</td><td>10</td></tr> <tr><td>2022</td><td>5</td></tr> <tr><td>2023</td><td>2</td></tr> <tr><td>2024</td><td>1</td></tr> <tr><td>2025</td><td>1</td></tr> </tbody> </table> Top 20 tags by frequency <table border="1"> <caption>Top 20 tags by frequency</caption> <thead> <tr> <th>Tag</th> <th>Count</th> </tr> </thead> <tbody> <tr><td>machine-learning</td><td>626</td></tr> <tr><td>deep-learning</td><td>193</td></tr> <tr><td>classification</td><td>175</td></tr> <tr><td>neural-network</td><td>172</td></tr> <tr><td>neural-networks</td><td>169</td></tr> <tr><td>python</td><td>148</td></tr> <tr><td>cross-validation</td><td>146</td></tr> <tr><td>regression</td><td>136</td></tr> <tr><td>data-mining</td><td>136</td></tr> <tr><td>svm</td><td>134</td></tr> <tr><td>random-forest</td><td>130</td></tr> <tr><td>optimization</td><td>130</td></tr> <tr><td>scikit-learn</td><td>124</td></tr> <tr><td>keras</td><td>121</td></tr> <tr><td>gradient-descent</td><td>120</td></tr> <tr><td>boosting</td><td>120</td></tr> <tr><td>r</td><td>119</td></tr> <tr><td>dataset</td><td>119</td></tr> <tr><td>loss-functions</td><td>118</td></tr> <tr><td>convolutional-neural-network</td><td>118</td></tr> </tbody> </table>	Year	Count	2010	10	2011	30	2012	40	2013	35	2014	75	2015	75	2016	93	2017	90	2018	90	2019	35	2020	30	2021	10	2022	5	2023	2	2024	1	2025	1	Tag	Count	machine-learning	626	deep-learning	193	classification	175	neural-network	172	neural-networks	169	python	148	cross-validation	146	regression	136	data-mining	136	svm	134	random-forest	130	optimization	130	scikit-learn	124	keras	121	gradient-descent	120	boosting	120	r	119	dataset	119	loss-functions	118	convolutional-neural-network	118
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Links to Relevant Documents

- Insert the links to any transcripts, recordings, survey results, or observation notes:

– (Link Here)

<https://drive.google.com/file/d/1k6wSzzI8QDEazg8m1Dx1s5B7CyCQiwqF/view?usp=sharing>

Question.Id	Title	Score	View.Count	Answer.Count	Is.Answered	Accepted.Answer.Id	Tags	Own
1	15135 Train/Test/Validation Set Splitting in Sklearn	206	459646	18	True	15136	machine-learning, scikit-learn, cross-validation	H
2	14899 How to draw Deep learning network architecture diag...	202	395897	7	True	130354	machine-learning, neural-network, deep-learning, sv...	M
3	5706 What is the "dyng ReLU" problem in neur...	201	168356	6	True	5734	machine-learning, neural-network, deep-learning	te
6	694 Best python library for neural networks	150	127062	17	True	695	machine-learning, python, neural-network	rr
8	410 Choosing a learning rate	116	130008	11	True	414	machine-learning, neural-network, deep-learning, op...	rz
9	10188 Why do cost functions use the square error?	115	72482	6	True	10192	machine-learning, linear-regression, loss-function	G
11	806 Advantages of AUC vs standard accuracy	100	110342	4	True	807	machine-learning, accuracy	ai
12	24452 In supervised learning, why is it bad to have correlate...	90	131702	7	True	24453	machine-learning, correlation	Si
13	5226 strings as features in decision tree/random forest	89	156658	6	True	5229	machine-learning, python, scikit-learn, random-fores...	u
14	16904 GBM vs XGBOOST? Key differences?	83	129567	5	True	16908	machine-learning, algorithms, xgboost, ensemble-m...	A
16	20296 Cross-entropy loss explanation	80	160004	6	True	20301	machine-learning, neural-network, deep-learning, so...	ei
19	9832 What is the Q function and what is the V function in r...	71	94410	7	True	10358	machine-learning, reinforcement-learning	M
20	16807 Why mini batch size is better than one single "b...	71	83746	5	True	16818	machine-learning, deep-learning, mini-batch-gradie...	H
22	24511 Why should the data be shuffled for machine learning...	71	108687	11	True	24539	machine-learning, neural-network, deep-learning	G
23	23159 In softmax classifier, why use exp function to do nor...	69	33697	5	True	24112	machine-learning, deep-learning	H
24	761 Clustering geo location coordinates (lat,long pairs)	67	108583	9	True	764	machine-learning, python, clustering, k-means, geos...	rc
25	45165 How to get accuracy, F1, precision and recall, for a ke...	66	208658	5	True	45166	machine-learning, neural-network, deep-learning, cl...	Z
26	2368 Machine learning - features engineering from date/ti...	63	68248	10	True	2370	machine-learning, time-series, feature-selection	Ic
27	12761 Should a model be re-trained if new observations are ...	63	68978	6	True	12763	machine-learning, predictive-modeling, optimization...	n
28	20179 What is the advantage of keeping batch size a power ...	63	50128	4	True	20193	machine-learning, training	Ja
31	9818 Is there any domain where Bayesian Networks outperf...	62	19853	9	True	9820	machine-learning, pgm	M
32	11619 RNN vs CNN at a high level	62	75869	5	True	11621	machine-learning, neural-network, beginner	Li
33	18583 LeakyReLU vs PReLU	62	62845	3	True	18589	machine-learning, deep-learning, neural-network, ac...	M

2–3 Paragraph Research Memo

• Summary and initial findings

Since 2016, the number of active members in this community has experienced a decrease. Especially from 2021 to 2024, there was a sharp drop in the number of individuals who asked questions in the Stack Exchange public forum. It is suspected that this occurred as a result of the invention of artificial intelligence, such as ChatGPT and other types of AI that can answer community questions much more quickly than responses in the Stack Exchange forum. On the other hand, it turns out that machine learning became the most widely discussed topic in the data science community, surpassing other subtopics such as neural networks, deep learning, and random forests.

Dataset Dictionary

Column Name	Description
Question Id	Unique identifier for the question
Title	Title of the question
Score	Net vote score (upvotes - downvotes)
View Count	Number of times the question has been viewed
Answer Count	Total number of answers posted
Is Answered	Boolean (TRUE/FALSE) indicating whether the question has at least one answer
Accepted Answer Id	ID of the accepted answer (if one exists)
Tags	List of tags associated with the question
Owner Display Name	Username of the question author
Owner Reputation	Reputation score of the question author
Creation Date	Date and time when the question was created
Last Activity Date	Most recent activity date (edit, answer, comment, etc.)
Link	URL link to the question
Site	Stack Exchange site name (e.g., StackOverflow, CrossValidated, etc.)

Zoomed-In Views of Figures 1 and 2

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Figure 1: Top 5 by Score

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Figure 2: Top 5 by Date